Contents

Traffic **1**
- Purpose . . . 1
- Role in the Testbed . . . 1
- Objectives . . . 1
- Traffic Analysis . . . 1
- Recommended Test Procedure . . . 2
- Safety . . . 2

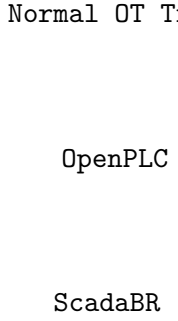
Traffic

Purpose

The `traffic/` component generates or manages normal OT network traffic for the laboratory environment. Its primary purpose is to establish realistic background Modbus/TCP communication against the simulated PLC.

Role in the Testbed

Normal traffic provides the baseline against which controlled abnormal activity can be compared.



The attacker component can operate alongside this baseline so that packet captures contain both normal and test traffic.

Objectives

- Generate repeatable Modbus/TCP communication.
- Keep the PLC exercised during experiments.
- Provide a realistic background for packet analysis.
- Make it possible to compare normal and abnormal Modbus operations.

Traffic Analysis

Wireshark can be used to inspect:

- TCP sessions
- Modbus/TCP requests and responses
- Function codes
- Register addresses

Generated by deepaj-2 on 2026-10-24 10:15:20 UTC

- Values
- Request timing
- Differences between normal and controlled write traffic

Recommended Test Procedure

1. Start the OT environment.
2. Allow normal traffic to run.
3. Capture traffic with Wireshark.
4. Establish a baseline.
5. Run a controlled security test.
6. Compare the resulting packets with baseline traffic.
7. Record observations.

Safety

Traffic generation should target only the simulated PLC inside the laboratory network.

Generated by deepaj-2 on 2026-10-24 10:15:20 UTC